Rushil Singh

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Education

Carnegie Mellon University Pittsburgh, PA
Bachelor of Science in Information Systems, Minor in Artificial Intelligence Expected May 2028
Relevant Coursework: Data Structures & Algorithms, Database Design & Development, Natural Language Processing, Designing Human-Centered Software, Discrete Math, Probability Theory, Linear Algebra, Calculus 3

Professional Experience

Perforated AI Pittsburgh, PA
Machine Learning Research Collaborator *May 2026 – October 2026*
- Co-authored *"Pruning Then Perforating: Dendrites Recover What Structured Pruning Removes"* (under ICLR 2027 review), investigating neural network compression by combining structured pruning with Perforated Backpropagation
- Evaluated L2 magnitude and Taylor-based pruning experiments on ResNet-18 image classification, improving accuracy by up to 1.60 percentage points through introducing artificial dendrite architectures to recover lost model accuracy

ListenFirst New York City, NY
AI Engineering Intern *July 2026 – August 2026*
- Built AI-powered reporting workflows as the Partnerships unit's sole intern, deploying organization-wide Claude Skills using Python, JavaScript, and RESTful APIs that reduced manual effort by 75% across 60+ projects for the team
- Developed production automation for monthly billing and scheduling across 40+ client accounts, cutting API call volume by 90%+ through streamlined data pipelines and surfacing $15K+ in billing discrepancies missed manually

Twinly Remote, EST
Software Engineer *June 2026 – August 2026*
- Built and shipped cross-platform AI agent capabilities in Swift and Python, developing voice cloning, reservation booking, and automated gaming features accessible through natural prompts for 100+ users and 2,000+ followers
- Contributed through code reviews and PR/MR-based feature integration while deploying Electron desktop builds across macOS and Windows and resolving platform-specific issues to maintain consistently updated releases

Activities & Involvement

180 Degrees Consulting Pittsburgh, PA
Student Consultant *February 2026 - Present*
- Analyze annual visitation, sales, and income data for Riverlife to identify demand patterns and rental opportunities, translating data-driven findings into recommendations for target customer segments, timings, and seasonal revenue

Traders @ CMU Pittsburgh, PA
Sports Markets Team Lead *February 2026 - Present*
- Lead a team investigating sports prediction-market arbitrage and edge cases, integrating live game data and APIs into Python research pipelines to analyze 15K Kalshi markets and evaluate low latency trading strategies after fees

Projects

SquidSpell – ASL Fingerspelling Tool | *Python, scikit-learn, MediaPipe, FastAPI, WebSockets, React, Supabase* | Live Demo
- Built a real-time Computer Vision app that runs browser hand tracking to recognize the ASL alphabet with MediaPipe and streams data over WebSockets to a Python service that classifies each letter at 99.4% test accuracy
- Deployed the full stack on Vercel and Render with 200+ automated tests, using Supabase (PostgreSQL, RLS, Google OAuth) for accounts, saved history, and a global speed-test leaderboard, with all game scoring handled on server-side

Snake Duel – Real-Time Multiplayer Game | *C++, WebAssembly, SDL2, WebSockets, Docker, CMake* | Live Demo
- Created a real-time online and local 1v1 multiplayer game in C++, sharing one game-logic core across a desktop app, a WebAssembly browser build, and the match server so every device remains synced identically with the server
- Shipped a 10 Hz authoritative WebSocket game server with an in-memory room registry and a JSON protocol, deployed as a single Docker container, with 73 automated tests spanning the simulation, bot AI, and wire protocol

Technical Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript, Swift, SQL, HTML, CSS
Frameworks & Libraries: PyTorch, React, Node.js, Flask, Pandas, NumPy, Matplotlib, Framer Motion, Tailwind CSS
Developer Tools: Git, GitHub, Linux, Docker, GitHub Actions, Supabase, PostgreSQL, Jupyter, Claude Code